

Referee's report on:
Space-time fractional stochastic heat equation with spatially
inhomogeneous white noise
by
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In this paper, the author studies the following space-time fractional stochastic heat equation:

$$\partial_t^\beta u(t, x) = -\nu(-\Delta)^{\frac{\alpha}{2}} u(t, x) + I_t^{1-\beta} \left(\lambda \sigma(u(t, x)) \dot{F}_\rho(t, x) \right), \quad t > 0, x \in \mathbb{R},$$

where $\beta \in (0, 1)$, $\nu > 0$, $\alpha \in (1, 2]$ and $\lambda > 0$. The main novelty of the paper is the study of the spatially inhomogeneous white noise $\dot{F}_\rho$; see (1) below. There are many grammar/spelling issues in the paper. We list some of those up to page 9. The author are highly recommended to go over the paper again to check out all similar issues. In terms of mathematics, the results seem not complete and there are also some issues in the proofs of the main results; see below. We recommend a **major revision** of the paper.

Suggestions and concerns:

1. The noise is introduced in Definition 1. It would be better to explicitly write out

$$\mathbb{E} \left(\dot{F}_\rho(t, x) \dot{F}_\rho(s, y) \right) = \delta_0(t - s) \rho(x - y) \delta_0(x - y). \quad (1)$$

2. At the end of Section 2.1, can you give some interesting, important, nontrivial examples of ρ ? Since this is the main contribution of the paper, one wants to see why such extension is necessary and important and how different properties it brings.
3. Theorem 1: It seems that the conditions for α and β are not optimal. The requirement on $\alpha \in (1, 2]$ can be relaxed. Indeed, by [1], the solvability is done under the following condition:

$$d < 2\alpha + \frac{\alpha}{\beta} \min(2\gamma - 1, 0)$$

By setting $d = 1$ and $\gamma = 1 - \beta$ for this paper, we see that

$$1 < 2\alpha + \frac{\alpha}{\beta} \min(1 - 2\beta, 0),$$

or equivalently,

$$\alpha > \frac{1}{2} \vee \beta.$$

So it seems the right condition on α should be

$$\alpha \in (1/2 \vee \beta, 2].$$

Moreover, I don't see why you cannot let $\beta = 1$ in your paper. Can you explain this point?

4. Remark 2, point No. 2: When $\rho \equiv c$, this reduces to the standard space-time white noise. All relevant statements in Theorem 1 has been also proved in [1]. Please make the comparison to see if the moment exponents in Eq. (6) and Hölder exponents you obtained are consistent with the results in [1].
5. In the third display on p. 13, the author uses

$$\frac{\partial}{\partial t} G_t(x - y) \leq \frac{C}{t} G_t(x - y) \quad (2)$$

to prove the time increments. First of all, I am not totally sure of this inequality, when one refers to [1], please give the precise equation or theorem so that the reader can easily local this result. Secondly, the left-hand side of (2) is signed while the right-hand side is nonnegative. But when applying for this inequality, one needs to take an absolute value. Would this cause any issue? In other words, one may want the following bounds instead:

$$\left| \frac{\partial}{\partial t} G_t(x - y) \right| \leq \frac{C}{t} G_t(x - y)$$

This kind of inequality does not seem correct. One may try the standard heat kernel $G_t(x) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}$. In this case, we see that

$$\frac{\partial}{\partial t} G_t(x) = \frac{1}{2} \left(\frac{x^2}{t} - 1 \right) \frac{G_t(x)}{t}.$$

One cannot find a constant C that is uniformly in t, x so that (2) is true. Even this paper handles the case $\beta < 1$, I don't expect much difference there.

Minor issues are listed below:

1. Line 3 of Section 2.1, “Randon measure” should be “Radon measure”. This misspelling appeared many times in the paper.

2. Line 1 on page 2: “satisfying” → “satisfies”.
3. Line 3 on third paragraph of page 2: Insert “the” in front of “STF-SHE”. Update this grammar issue for the whole paper.
4. Line 4 on page 3: “grows” → “grow”.
5. Line 8 on page 3: “in bounded domain” → “in a bounded domain”.
6. Line -15 on page 3: “SPDE” → “SPDEs”.
7. Line -12 on page 3: Insert “the” in front of “moments”.
8. Line -1 on page 3: “the Green function” → “the Green’s function”. Update the whole paper for this issue.
9. Line 1 of Remark 1: “the author in [27]” → “Zähle [27]”.
10. Line 10 on page 6: “field” → “fields”.
11. Title of Section 3: “Moments bounds” → “Moment bounds”.
12. First paragraph of both Section 3 and Section 3.1: “some Banach space” → “the Banach space”.
13. Line -2 in the first paragraph of Section 3: “the solution $u(t, x)$ grow at least exponentially” → “the moments of the solution $u(t, x)$ grow at least exponentially”.
14. Line 2 in the statement of Theorem 1: “satisfying” → “satisfies”.
15. Remark 2 Point 1: “The order of fractional operator” → “The order of the fractional operator”.
16. Line 1 in Point 2 of Remark 2: Insert “then” in front of “one”.
17. Line -4 on page 8: “reduces” → “reduce”.
18. Line -3 on page 8: “coincides the results” → “coincides with the results”.
19. Line -1 on page 8: “is” → “are”.
20. Line 1 on page 9: “order” → “orders”.
21. Line 2 on page 9: “studying stochastic heat” → “studying the stochastic heat”.

References

- [1] Chen, Le, Yaozhong Hu and David Nualart. (2019) Nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$. *Stochastic Processes and their Applications*, 129, 5073–5112.